

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		award (1) for either of following - the aldehyde is being oxidised to a carboxylic acid and reduced to an alcohol - the aldehyde gains oxygen forming a carboxylic acid and gains hydrogen forming an alcohol			1	1		
		(ii)		moles of C ₆ H ₅ CHO = $\frac{9.5}{106.06}$ = 0.0896 (1) this should make 0.0448 mol of C ₆ H ₅ CH ₂ OH (1) percentage yield = $\frac{3.39}{4.84} \times 100$ = 70 % (1) accept alternative method moles of C ₆ H ₅ CH ₂ OH = $\frac{3.39}{106.08}$ = 0.0314 (1) formed from 2 × 0.0314 = 0.0628 mol of C ₆ H ₅ CHO (1) mass of C ₆ H ₅ CHO = 6.66 g percentage yield = $\frac{6.66}{9.50} \times 100$ = 70 % (1)		3		3	2	
		(iii)		2C ₆ H ₅ CHO + NaOH → C ₆ H ₅ CH ₂ OH + C ₆ H ₅ COONa			1	1		
				Question 10 total	6	3	2	11	2	0